

Quiz - 1 (Basics of Electronics Engineering - EF)

Total points 26/26 ?

The respondent's email (redacted) was recorded on submission of this form.

0 of 0 points

Name *

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Admission Number *

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Mobile Number *

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Quiz - 1 (Basics of Electronics Engineering - EF)

26 of 26 points

There are 21 questions in total



✓ Dynamic resistance of a diode is –

1/1

- ☐ Directly proportional to the diode current
- ☐ Does not depend on the diode current
- ☒ Inversely proportional to the diode current
- ☐ Independent of the operating temperature



✓ The rating of Electrolytic capacitor is measured in –

1/1

- ☐ Farad
- ☐ Volt
- ☐ Neither Volt nor Farad
- ☒ Both (a) and (b)



✓ A power supply is delivering 100 W to a load of 10k Ω . What is the AC (ripple) voltage across the load if the ripple factor is 0.1%?

2/2

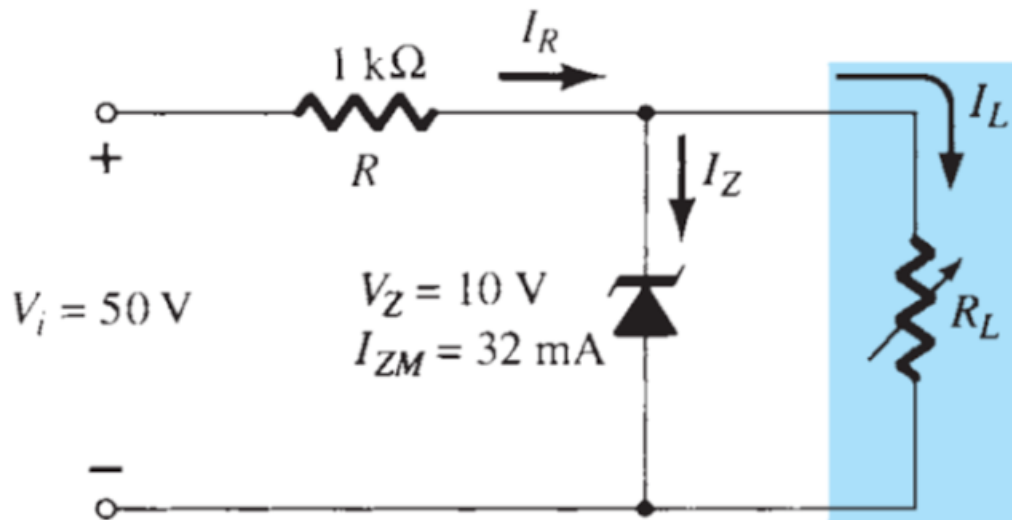
- ☒ 1 V
- ☐ 100 V
- ☐ 0.1 V
- ☐ 10 V





1/1

Determine the maximum value of the R_L in the following circuit.



- ☒ 1.25 kΩ
☐ 0.25 kΩ
☐ Any value over 0.25
☐ Cannot be determined based on the given data



✓ A Half Wave Rectifier uses a transformer of turn ratio 2:1 and a load resistance of 500Ω. The source and the forward resistance of the diodes are negligible. If the rms primary voltage is 240V, the DC output voltage is –

2/2

- ☒ 54 V
☐ 76 V
☐ 108 V
☐ 38 V



✓ For 180 kilo ohm resistor with 10% tolerance, the colour bands are in the sequence of 1/1

- ☒ brown-grey-yellow-silver
- ☐ brown-grey-red-silver
- ☐ brown-grey-red-gold
- ☐ brown-grey-yellow-gold



✓ While checking a capacitor with multi-meter if the pointer deflects and remain there that means - 1/1

- ☐ The capacitor is Open
- ☒ The capacitor is Short
- ☐ The capacitor is intact
- ☐ None of these



✓ In the depletion region of a p-n junction, there is a shortage of - 1/1

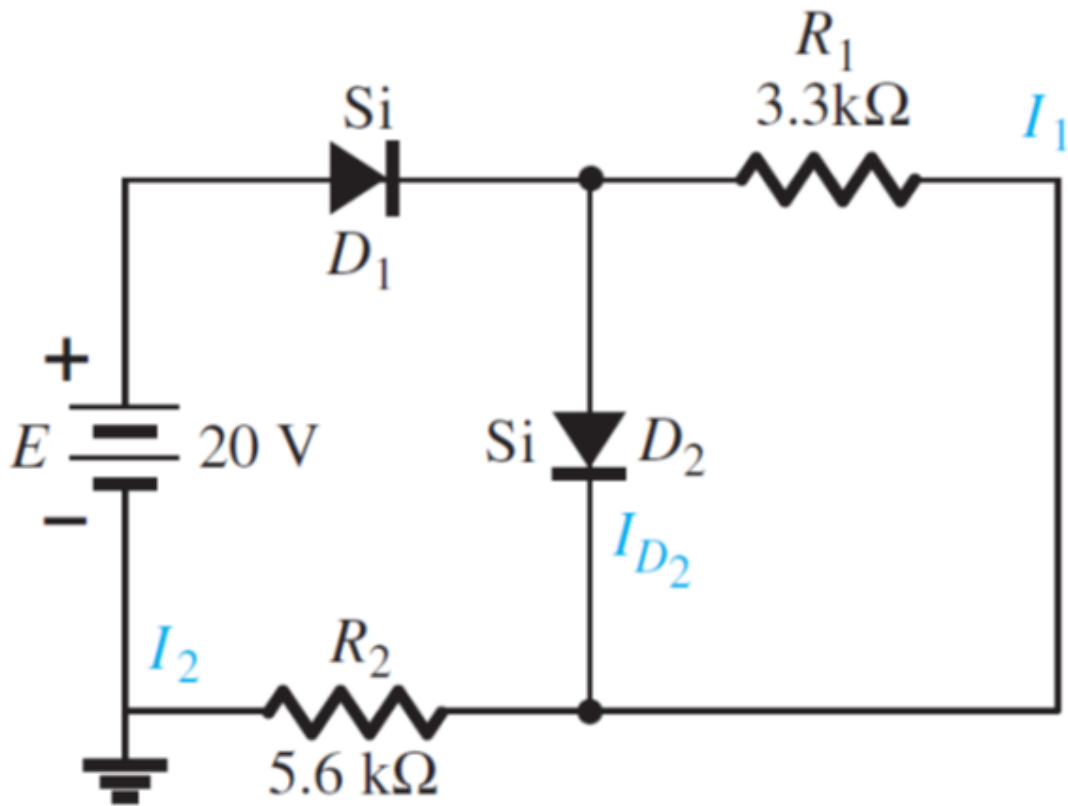
- ☐ donor ions
- ☐ none of the other options are true
- ☐ acceptor ions
- ☒ holes and electrons





2/2

Determine I_{D_2} in the following circuit.



- ☐ Can not be determined
- ☐ 3.22 mA
- ☒ 3.11 mA
- ☐ 3.33 mA



A BJT is a –

1/1

- ☐ current controlled voltage device
- ☐ voltage controlled voltage device
- ☒ current controlled current device
- ☐ voltage controlled current device



✓ If the current flowing through a p-n junction diode increases 10 times, what is the approximate increase in diode voltage? Assume forward biased Si diode operates at room temperature. 2/2

- ☐ 120 mV
- ☐ 30 mV
- ☐ 90 mV
- ☒ 60 mV



✓ Reverse saturation current – 1/1

- ☒ None of the other options are true
- ☐ Does not depend on temperature
- ☐ Almost doubles per 7°C rise in temperature
- ☐ Almost doubles per 5°C rise in temperature

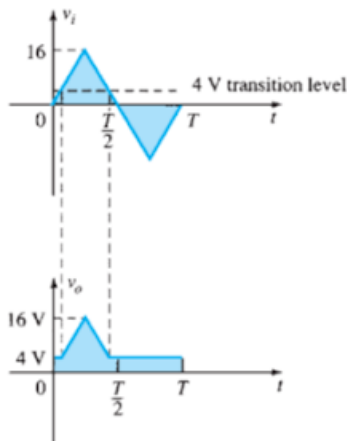


✓ BJT is - 1/1

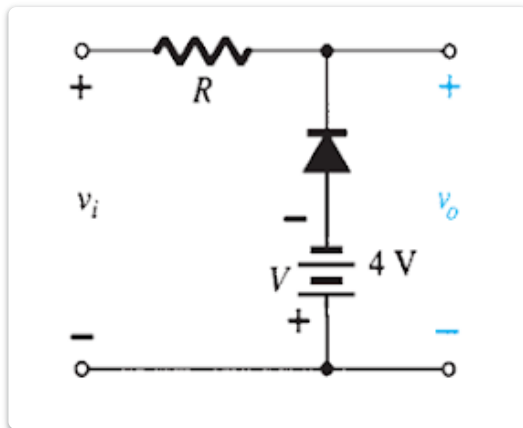
- ☐ A passive component
- ☒ An active component
- ☐ Both (a) and (b)
- ☐ None of these



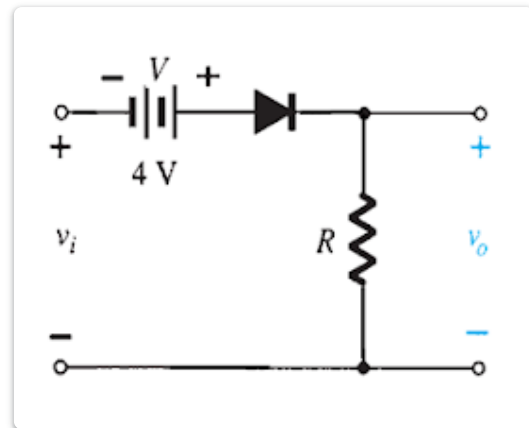
Input and output of a circuit is shown as follows.



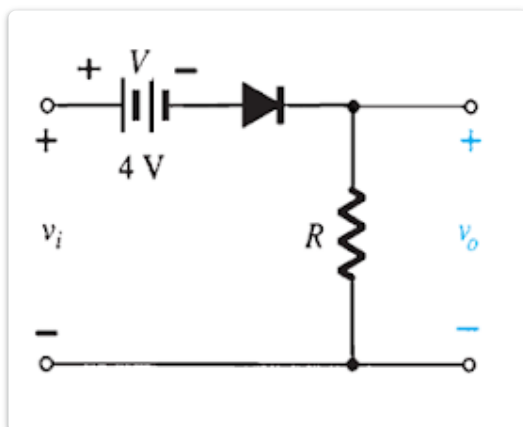
Which of the following circuit is responsible for producing the same?



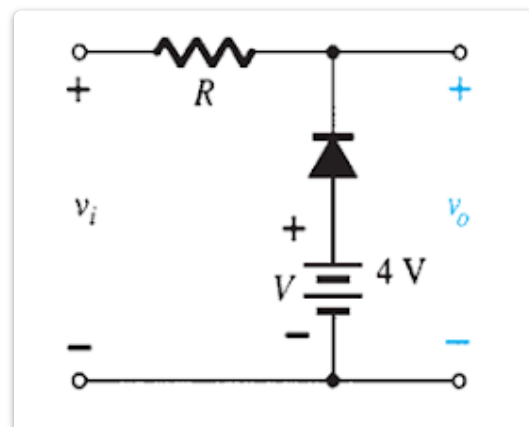
☐ Option 2



☐ Option 3



☐ Option 4



☒ Option 1



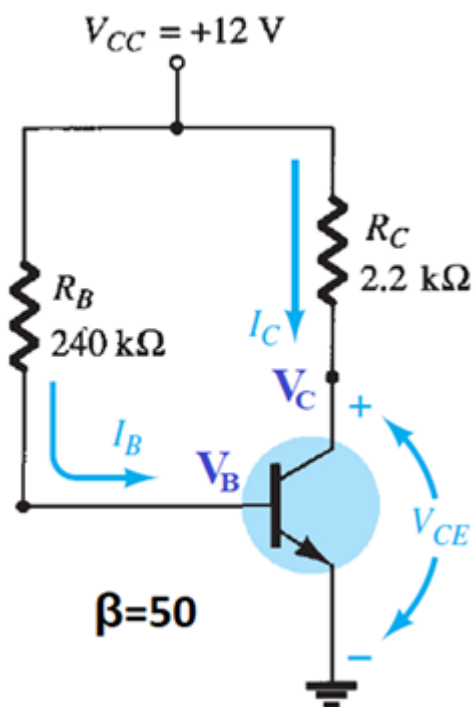
✓ If the doping to a pure semiconductor increases, the bulk resistance of the semiconductor - 1/1

- ☐ remains unchanged
- ☒ decreases ✓
- ☐ increases in p-type and decreases in n-type semiconductors
- ☐ increases

✓

2/2

Determine the V_{BC} in the following circuit.



- ☒ -6.13 V ✓
- ☐ None of the other options are correct
- ☐ 6.13V
- ☐ 6.83V

✓ If the input frequency of a Full Wave Rectifier is 50 Hz, the fundamental output frequency will be – 1/1

- ☒ 100Hz
- ☐ 200HZ
- ☐ 50 HZ
- ☐ 75HZ



✓ In a Si npn transistor, the V_{BE} and V_{CB} are 0.7 V and 0.2 V respectively. The transistor is operating in – 1/1

- ☐ inverse active region
- ☐ cut-off region
- ☐ saturation region
- ☒ active region



✓ The rating of resistors is measured in – 1/1

- ☐ ohms
- ☐ Watts
- ☐ amperes
- ☒ Both (a) and (b)



✓ An extrinsic semiconductors can be said as electrically –

1/1

- ☐ negative
- ☐ either positive or negative
- ☐ positive
- ☒ neither positive nor negative



✓ A capacitor has number code '103'. It's value is –

1/1

- ☒ 0.01 micro Farad
- ☐ 1.0 micro Farad
- ☐ 0.001 micro Farad
- ☐ 0.1 micro Farad



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